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When Generative AI Meets Socratic Method: Investigating Programming Learning Dynamics Through Behaviours, Interaction Qualities and Perceptions

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ABSTRACT

Background: The integration of generative artificial intelligence (GAI) tools like GPT into programming education offers transformative potential through personalised guidance and instant feedback, yet risks fostering overreliance and superficial learning due to their tendency to deliver direct, context-free answers.

Objectives: This quasi-experimental study addresses this gap by proposing a Socratic questioning framework to optimise GAI-facilitated programming instruction, emphasising critical thinking over passive solution retrieval.

Methods: We compared two pedagogical approaches: GAI-Scaffolded Learning (GSL), where GPT employs structured Socratic dialogue to guide problem-solving and GAI-Direct Learning (GDL), which provides immediate answers without guided inquiry. This research collected learners' programming behaviours, interactions data with GPT from screen recordings and platform log data and perceptions data. This research further utilised multiple learning analytics approaches (i.e., click stream analysis, lag-sequential analysis, epistemic network analysis [ENA] and statistics) to compare learners' programming behaviours, interaction patterns and perceptions under two approaches.

Results and Conclusions: Through an analysis of 80 college students' programming behaviours, interaction qualities and perceptions, we found some intriguing results. First, GSL engaged in cyclical, reflective practices (debugging, Socratic questioning, console use), while GDL prioritised rapid fixes via trial-and-error with GPT code, risking superficial mimicry and over-reliance on external resources. Second, ENA highlighted GSL's deeper engagement through interconnected feedback, emotional support and iterative inquiry, reducing frustration and sustaining persistence and GDL interactions focused on surface-level queries, lacking scaffolding for emotional/heuristic integration. Third, GSL maintained positive attitudes due to structured prompts aligning expectations and easing cognitive load. GDL attitudes declined from mismatched expectations and frustration.

Implications: Based on these findings, the study proposes pedagogical and developmental implications for future design and development of AI-augmented curricula, providing actionable insights for educators seeking to harness GAI's potential while nurturing critical thinking in programming education.

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Lay Summary

- What is currently known about this topic?
  - Generative AI provides instant feedback but risks student overreliance on code.
  - Direct AI answers often lead to superficial learning and lack of critical thinking.
  - Personalised guidance is vital for effective programming education.
  - Scaffolding is a proven method to help students solve complex problems.
- What does this paper add?
  - It introduces a Socratic questioning framework to guide AI-led instruction.
  - GSL students showed more reflective debugging habits than GDL students.
  - ENA analysis shows that guided AI dialogue reduces student frustration levels.
  - The study proves structured prompts better maintain positive student attitudes.
- Implications for practice and/or policy
  - Educators should design AI prompts that ask questions rather than give answers.
  - AI-augmented curricula must prioritise critical thinking over code generation.
  - Training should help students use AI as a tutor rather than a simple tool.
  - Policy should encourage AI tools that provide emotional and heuristic support.

1 | Introduction

In the modern era, programming education has emerged as a critical skill, driven by its pivotal role in fostering technological literacy and problem-solving capabilities (Sun, Boudouaia, Yang, et al. 2024). As industries increasingly rely on data-driven decision-making and advanced technologies, students equipped with programming skills gain a competitive edge in the job market, demonstrating adaptability and computational competence (Nouri et al. 2020). However, mastering programming is not without challenges, as learners often encounter obstacles that hinder their academic progress and confidence (Yilmaz and Karaoglan Yilmaz 2023).

The rise of generative AI (GAI) is fundamentally reshaping education, prompting scholars to define its potential roles (e.g., tutor, collaborator, tool), outline effective implementation strategies and identify key research issues that need to be addressed in this new era (Lai and Tu 2024). The integration of GAI has profoundly impacted programming education, making it more accessible and engaging. GAI-powered chatbots, particularly large language models like ChatGPT, are transforming the learning landscape by offering personalised guidance and real-time feedback (Andersen et al. 2022; Tian et al. 2023; Yilmaz and Karaoglan Yilmaz 2023b). Specifically for programming, these tools serve as a '24/7 tutor', assisting learners with a range of practical tasks from code generation and optimization to debugging, error analysis and explaining complex algorithmic logic (Kashefi and Mukerji 2023; Phung et al. 2023; Surameery and Shakor 2023). Through interactive dialogue, students can explore multiple solutions and deepen their understanding (Gayed et al. 2022; Ouh et al. 2023). Its ability to simulate human-like conversations makes it a versatile resource for students across diverse learning backgrounds, including those struggling with traditional pedagogical methods (Kohnke et al. 2023).

Despite its advantages, GAI tools' limitations pose significant challenges. The model generates responses based on pre-trained data without contextual memory, meaning each interaction is isolated and lacks continuity (Dwivedi et al. 2023; Gayed et al. 2022). Moreover, its tendency to present answers as definitive truths, without citations or nuanced qualifiers, risks fostering overreliance among students, who may accept outputs uncritically (Cooper 2023; Rawas 2024). The pedagogical effectiveness of GAI is heavily dependent on how students perceive and engage with the tool. Recent research shows that students' conceptions of ChatGPT-supported learning can vary widely (Tu and Hwang 2023) and a common risk is the tendency towards overreliance on the AI for direct answers, which can hinder the development of genuine problem-solving skills (Sun, Boudouaia, Yang, et al. 2024; Tian et al. 2023). This issue is, particularly, severe in programming education, where haphazard use of ChatGPT without structured guidance can lead to superficial understanding and flawed code implementation. For instance, students may input poorly formulated queries, assuming the tool will inherently produce accurate results, thereby limiting their ability to troubleshoot independently (Sun, Boudouaia, Yang, et al. 2024). Moreover, many GAI tools often provide learners with direct answers, which could undermine their ability to engage in self-directed problem-solving and critical thinking (Amos et al. 2025). This limitation highlights the need for a pedagogical framework that can guide GAI's capabilities towards fostering higher order thinking.

The Socratic teaching method, rooted in inquiry-driven dialogue, has been empirically shown to be effective in improving a range of student outcomes, including learning attitudes, higher-order thinking skills, self-reflection and overall performance (Dinkins and Cangelosi 2019; Lee 2014). It is inquiry-driven because it fundamentally shifts the cognitive locus of control to the student. The GAI's role is not to provide answers but to pose questions that stimulate the learner's own investigation into their code, assumptions and logic. This process transforms the learner from a passive information consumer into an active problem-solver. Building on this established pedagogical value, recent research has begun to explore the integration of Socratic principles into GAI to enhance both student learning and the reasoning capabilities of the models themselves (Chang 2024; Ma et al. 2024). To address the above-mentioned challenges, this study integrated Socratic questioning strategies into GAI to facilitate college students' programming learning. This study is grounded in a constructivist learning theory, which posits that learners actively construct their own knowledge and understanding through experience and interaction rather than passively receiving information (Vygotsky 1978). By designing GAL tools to deliver step-by-step, diagnostic feedback rather than immediate solutions, this approach aimed to cultivate critical thinking and deeper conceptual understanding. In detail, this research investigated the impact of Socratic questioning strategy on

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college students' programming behaviours, interaction qualities and perceptions when using GAI. This was achieved by comparing two kinds of classes: GAI-Scaffolded Learning (GSL) class refers to an AI-augmented pedagogical approach where GAI proactively employs Socratic questioning to stimulate thinking, contrasting with GAI-Direct Learning (GDL), which provides immediate answers without guided inquiry. As a result, this study attempted to address the following research questions:

RQ1. *How did the impact of GSL on learners' programming behaviors contrast with the effect of GDL?*

RQ2. *How did the impact of GSL on learners' interaction qualities with GAI contrast with the effect of GDL?*

RQ3. *How did the impact of GSL on learners' perceptions towards GAI contrast with the effect of GDL?*

2 | Literature Review

2.1 | Enhancing Programming Education Through AI-Powered Chatbots

A strong foundation in computer programming is critical for students to thrive as innovators in an increasingly digital world (Amos et al. 2025). Advances in AI, particularly in chatbot technology, are reshaping programming education by offering dynamic tools to accelerate skill development and pedagogical innovation. As AI continues to evolve, its integration into education underscores the growing necessity for students to master coding and digital competencies (Sun et al. 2022). Programming education, recognised for its practical applications, is now a cornerstone of AI-driven learning strategies (Liu et al. 2023). Recent research highlights the transformative potential of AI tools like chatbots, virtual instructors and interactive coding platforms in making programming education more accessible and engaging (Yilmaz and Karaoglan Yilmaz 2023b). These resources empower learners to grasp foundational programming concepts through real-time feedback and personalised guidance (Tian et al. 2023). For instance, AI-driven platforms enable students to generate readable code, identify errors efficiently and refine their problem-solving skills, fostering accelerated learning (Yilmaz and Karaoglan Yilmaz 2023). Moreover, such tools promote self-directed learning by allowing students to explore coding concepts independently, accommodating diverse learning paces and styles (Eason et al. 2023).

Among AI tools, ChatGPT from openai has emerged as a groundbreaking resource in programming education. Unlike conventional tools, ChatGPT engages learners through natural language interactions, offering tailored explanations and adaptive support (Dwivedi et al. 2023; Jeon and Lee 2023). Kim (2023) analysed ChatGPT-3.5's capabilities, highlighting ChatGPT-3.5's versatility in education, programming and content creation. Key strengths include personalised feedback, code debugging, business plan generation and creative content production, with interactive, context-aware responses setting it apart from traditional search engines. Studies demonstrate its versatility in tasks ranging from code generation and debugging to feedback provision and peer programming (Phung et al. 2023). For example, GPT-4 has shown proficiency comparable to human instructors in generating hints, grading assignments and contextualising complex concepts (Liu et al. 2025). Its intuitive interface democratises programming education, making it accessible even to novices (Surameery and Shakor 2023). ChatGPT's strengths lie in its ability to simplify complex topics using relatable analogies and real-world examples (Kohnke et al. 2023). Empirical studies illustrate its utility in practical scenarios, such as developing web-based tasks using JavaScript, HTML and CSS (Avila-Chauvet et al. 2023). While it cannot replace human programmers, ChatGPT significantly reduces coding time and expands creative problem-solving options (Biswas 2023). Its applications extend to code completion, error correction, optimization and technical documentation, making it invaluable for both learners and educators (Vukojičić and Krstić 2023). For students, ChatGPT serves as a 24/7 tutor, offering step-by-step guidance on debugging, syntax issues and algorithmic logic (Kashefi and Mukerji 2023). It also fosters deeper understanding through interactive dialogues, enabling learners to explore multiple solutions to programming challenges (Ouh et al. 2023). For instance, in undergraduate Java courses, ChatGPT has generated concise, well-structured code while highlighting memory-efficient alternatives—though its performance diminishes with tasks requiring intricate class integration or detailed explanations (Ouh et al. 2023).

Despite its advantages, GAI tools face challenges in handling highly complex or context-specific coding tasks, particularly regarding the reliability of generated outputs. Empirical evidence indicates that LLMs frequently generate hallucinatory outputs, including semantically invalid API invocations, logically inconsistent algorithms and non-executable code constructs that risk misleading novice programmers and impeding foundational skill development (Zhang et al. 2025). Educators must design assignments that balance AI assistance with authentic assessment to mitigate risks of overreliance or academic dishonesty (Tian et al. 2023). Additionally, while ChatGPT excels in generating code snippets and troubleshooting, its inability to fully replicate human intuition underscores the irreplaceable role of instructors in fostering critical thinking and creativity (Sun, Boudouaia, Yang, et al. 2024). The integration of AI tools like ChatGPT into curricula requires thoughtful implementation. Educators are encouraged to leverage these technologies to supplement traditional teaching methods. By combining AI's efficiency with human mentorship, institutions can cultivate adaptable learners prepared for evolving technological landscapes (Yilmaz and Karaoglan Yilmaz 2023; Amos et al. 2025). Future research should explore hybrid models that optimise GAI's strengths while addressing its limitations, ensuring equitable and effective programming education for all skill levels.

2.2 | Socratic Questioning Strategy Used in Education

The preceding analysis makes it clear that the mere presence of GAI is insufficient to guarantee deep learning. The literature

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highlights a critical challenge: without pedagogical guidance, learners risk becoming passive recipients of information, failing to develop the essential critical thinking and problem-solving skills that instructors aim to foster (Sun, Boudouaia, Yang, et al. 2024; Tian et al. 2023). This creates a distinct research gap and a practical need for instructional frameworks that can deliberately structure GAI interactions to promote active knowledge construction. One such framework, rooted in established learning science, is the Socratic method, which offers a powerful model for transforming GAI from a simple answer engine into a cognitive partner.

Socratic questioning originated in ancient Greece. Socrates guided others to identify contradictions or biases by questioning assumptions and engaging in progressive reasoning over 2400 years ago (Zare and Mukundan 2015). Through this continuous process of inquiry, he helped correct incomplete or inadequate beliefs, leading others to gradually achieve cognitive transformation and foster thinking. Philosopher Nelson (1949) argued that the uniqueness of Socratic dialogue lies in guiding learners to think and live like philosophers. By actively posing questions, seeking evidence and reasoning, learners come to understand the essence of things. This approach gradually evolved into the Socratic method of dialogue, giving rise to forms such as Socratic questioning and the Socratic teaching method. Gustav Heckmann (1988) distilled a dialogue-based teaching method from Socratic dialogue. Socrates inspired thought by continuously questioning students and through this process of being persistently questioned, students were eventually able to independently form new understandings and learn to regulate their thinking, closely relevant to the LLM prompting techniques (He et al. 2024). This two-way interaction and inquiry method prevented students from passively receiving information and instead 'forced' them to acquire knowledge through dialogue.

Socratic questioning essentially promotes individual cognitive development through structured inquiry, including clarification questions, probing assumptions, probing reasons and evidence, questioning viewpoints and perspectives, examining implications and consequences (Paul and Elder 2016). Socratic method has been shown to play a significant role in promoting students' construction of knowledge, conceptual acquire and the development of critical thinking (Goldin et al. 2017). Existing studies have integrated the Socratic method into GAI and conducted a series of empirical studies. Held et al. (2024) developed the AI therapist Socrates based on multiagent LLM tool, which receives automated feedback from an AI supervisor and AI rater. The study gathered qualitative and quantitative data about users' and clinicians' experience of interacting with the tool, the goal is to inform the eventual implementation of GAI tools. Results indicated the development of Socrates are important first steps in applying GAI to mental health treatment delivery. In addition, Khan Academy has developed the Socratic tutor Khanmigo based on GPT-4, which encourages students to engage in deep thinking rather than providing direct answers, fostering their critical thinking skills (Khan Academy 2024). Etkin et al. (2025) determined the impact of four GPT-based tools on college-aged participants' reading comprehension of standardised American College Test-derived passage. Results showed that low performers were most benefited by the Socratic chatbot. Although certain applications of GPT may undermine students' critical thinking and comprehension (Vargas-Murillo et al. 2023), studies have demonstrated that the integration of Socratic questioning within this tool can significantly enhance both critical thinking and comprehension among students (Mahmud and Tryana 2023). Chang (2024) developed the SocraSynth prompt template for Socratic questioning reasoning using multiple agents, inspiring human reasoning through debates and collaboration between the agents. Ma et al. (2024) investigated the integration of Socratic teaching methods in GAI chatbots and its impact on enhancing student learning experiences and outcomes. The research aimed to address the current limitations of AI chatbots in educational settings, such as the risk of students' over-reliance on direct answers, by fostering deep thinking and active learning. Constructivist theory provides a robust framework for understanding how technology can support learning. Central to this framework is Vygotsky's (1978) concept of the Zone of Proximal Development (ZPD), which describes the gap between what a learner can achieve independently and what they can achieve with guidance. The process of providing temporary, tailored support to bridge this gap is known as scaffolding. Effective scaffolding makes learning manageable by reducing cognitive load and modelling expert processes and it is gradually removed as the learner's competence grows. Integrating the Socratic method into GAI involves not providing direct answers, but rather posing structured questions to guide learners along a path of reflective reasoning. Learners seek evidence and engage in logical reasoning to develop solutions to problems. This approach offers personalised guidance and real-time feedback, stimulating their curiosity and fostering critical thinking.

The concept of a computational system acting as a Socratic tutor is not new; it is a cornerstone of the field of Intelligent Tutoring Systems (ITS). For decades, ITS research has demonstrated that learning can be significantly enhanced when a student interacts with a system that provides personalised, step-by-step guidance (Mousavinasab et al. 2021). Classic ITS, such as AutoTutor or Cognitive Tutors, achieve this through a sophisticated architecture comprising a domain model (expert knowledge), a student model (tracking learner progress and misconceptions) and a pedagogical model that uses this information to deliver rule-based feedback or Socratic hints (Xu et al. 2019). However, the advent of large-scale GAI like GPT-4 introduces a fundamentally different paradigm, creating a clear research gap. While traditional ITS rely on preprogrammed pedagogical logic and explicit student models, GAI operates without either. Its ability to engage in dialogue is not derived from a structured database of expert knowledge but from patterns learned across vast datasets, making its pedagogical capabilities emergent rather than programmed. This distinction is critical. The high cost and domain-specificity of developing traditional ITS have limited their widespread adoption. GAI, in contrast, offers a domain-general, highly flexible and accessible alternative. The central research gaps this study addresses is whether carefully engineered prompts can guide a GAI to effectively simulate the role of a Socratic tutor, signalling a paradigm shift from structured, model-driven systems to flexible, language-driven cognitive partners. Our study is therefore designed to provide initial evidence in this

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TABLE 1 | Participant's information.

Group	Number	Average age	Condition
Experimental group	36 (19 females, 17 males)	21.5	GAI-Scaffolded Learning
Control group	44 (24 females, 20 males)	21.5	GAI-Direct Learning

nascent area, examining how a GAI, when prompted to adopt a Socratic stance, influences student learning processes and outcomes in programming education.

3 | Research Methodology

3.1 | Research Context, Participants and Instructional Procedures

This research was carried out within a compulsory 'Python Programming' course for Educational Technology students at a Chinese university during the Fall 2024 semester. Using a quasi-experimental approach, the study involved 80 undergraduate students (see Table 1) randomly assigned to one of two conditions with an average age of approximately 21.5 years: an experimental group ($n = 36$; 19 females, 17 males) and a control group ($n = 44$; 24 females, 20 males). The core distinction between the groups lay in the pedagogical structure of their interaction with a GAI. The experimental group was assigned to a GSL condition. This condition implemented a pedagogical intervention where the GAI acted as a Socratic tutor. Before revealing a full solution, the GAI engaged students in a guided dialogue, posing questions to help them reflect on their errors, break down the problem and construct the solution more independently (see Figure 1a). The focus was on scaffolding the problem-solving process itself. In contrast, the control group was assigned to a GDL condition. This condition was designed to simulate the common use of GAI as a direct answer engine. When students requested help, the GAI provided immediate and complete code solutions without intermediate steps (see Figure 1b). The focus was on direct problem resolution. Both groups were instructed by the same educator (the study's first author), who ensured uniformity in teaching methodology across conditions. Participants had prior C programming experience and ethical approval for data collection and student engagement was secured through formal institutional review.

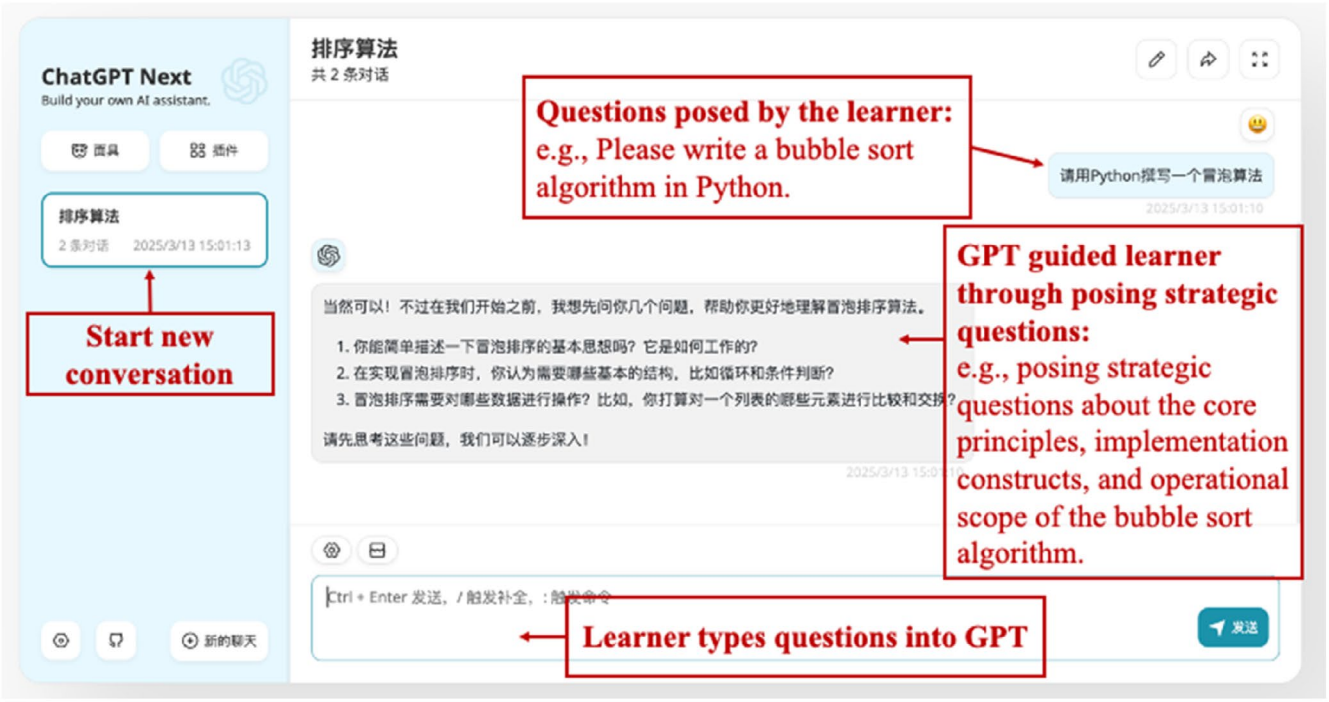
Instructional content and pacing remained identical for both groups, except for Socratic questioning techniques integrated exclusively into the GSL condition. The course spanned two phases over six 90-min sessions. Phases I (Sessions 1–4) introduced foundational Python concepts, including syntax basics (e.g., IDLE, input/output functions), data types (integers, lists, dictionaries), control structures (conditionals, loops) and function design (recursion, lambda expressions). Phase II focused on collaborative Python project development using GPT support. The instructional sessions were structured in alignment with the Python Programming textbook (ICOURSE 2024). A custom-developed platform, GPT Next (adapted by the first author), served as the primary interface for student problem-solving (refer to Figure 1). This platform integrated a GPT-4 model, allowing learners to pose questions and engage in topic-specific dialogues with the GAI to receive tailored feedback. While the GSL group experienced Socratic questioning from the GAI before accessing full code answers, the GDL group interacted with GPT-4 to obtain direct code solutions. During Phases I, both cohorts began sessions with instructor-led lectures, multimedia presentations and live Python demonstrations, followed by independent coding exercises. Phase II required students to design and implement a programming project within a 90-min timeframe, leveraging GPT-4 assistance as needed.

3.2 | Design of the Socratic-Questioning

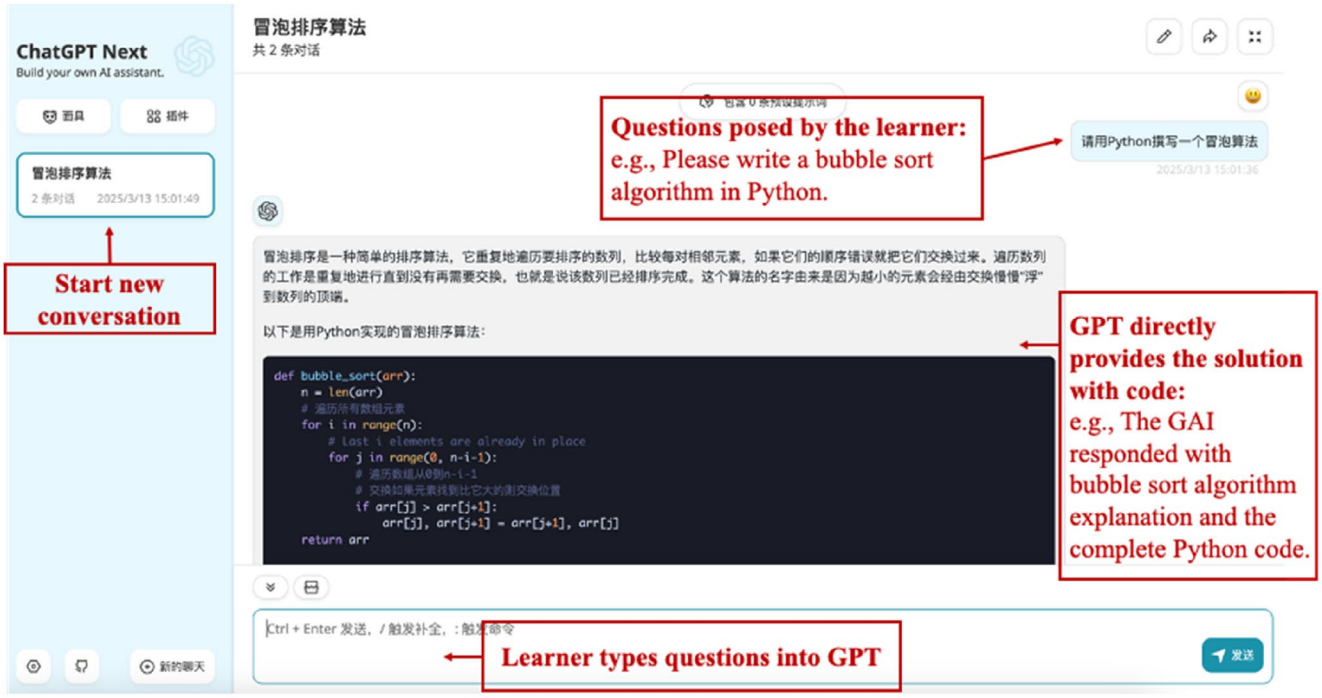
A core feature of the GSL condition was its Socratic-questioning framework. The questions were not pre-scripted but were dynamically generated in real time by the GPT-4 model. To ensure rigour and pedagogical soundness, the model's behaviour was governed by a detailed system prompt grounded in a dual-foundation design.

First, the framework was adapted from OpenAI's 'Steerability: Socratic tutor' guidelines (OpenAI 2024), providing a robust conversational structure. Second, it integrated implications from recent peer-reviewed studies on GAI applications in programming education (Sun, Boudouaia, Yang, et al. 2024; Sun, Boudouaia, Zhu, et al. 2024; Yilmaz and Karaoglan Yilmaz 2023). This ensured that the guidance was not generic but specifically tailored to the challenges and learning opportunities in the programming domain, prioritising iterative inquiry over direct solutions. Based on this synthesis, we, the researchers, refined the framework for the context of self-driven projects. The principle of 'never giving the answer' was modified to: 'you will provide the answer only once students demonstrate a clear understanding of the problem through prior dialogue'. Operationally, we defined 'a clear understanding' as the student's ability to articulate both the root cause of the problem and the logical steps required for its solution. This was implemented by instructing the AI system to analyse student responses for two key indicators: (1) accurate identification of the underlying issue, such as distinguishing a logical error from a syntax error or pinpointing a conceptual misunderstanding like variable scope and (2) a coherent explanation of the solution path, such as outlining step-by-step logic involving initialization, iteration and conditional checks. The AI was then prompted to use its inherent reasoning capacity to make a holistic judgement based on these criteria before deciding to provide a direct answer. This adjustment, informed by the literature, was intended to balance cognitive scaffolding with the need to prevent frustration and sustain motivation. The reliability of this method was established by applying the identical, standardised system prompt to all GSL participants, ensuring a consistent pedagogical strategy. The validity of the framework is strongly supported by its design process. Grounding the prompt

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(a)



(b)

FIGURE 1 | GPT Next platform used in class. (a) GAI-Scaffolded Learning group (GSL) and (b) GAI-Direct Learning group (GDL).

in both industry best practices and relevant academic research enhances its content validity. Furthermore, the prompt's instructions explicitly operationalise the theoretical construct of Socratic scaffolding, supporting its construct validity. For full transparency, the detailed Socratic-questioning prompt used in the GSL group is shown in Figure 2 and the basic prompt for the GDL group is in Figure 3.

3.3 | Data Collection and Analysis Approaches

Three primary methods were utilised to collect and analyse data in this study (see Table 2). First, student programming behaviours were recorded through muted screen recordings. For our primary analysis, we selected the final experimental session as the focal video dataset. This session, scheduled for 40 min,

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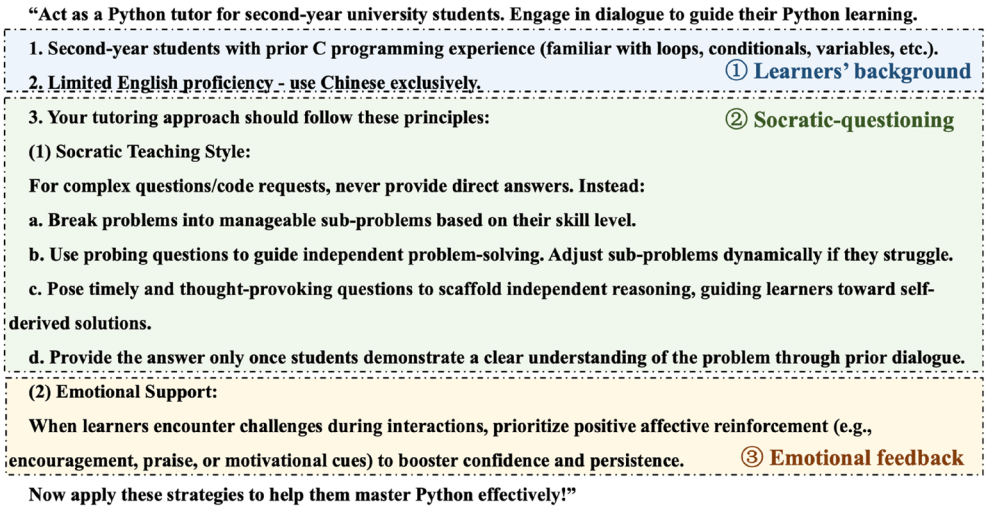


FIGURE 2 | Prompt design illustrating the core components of the Socratic-questioning strategy.

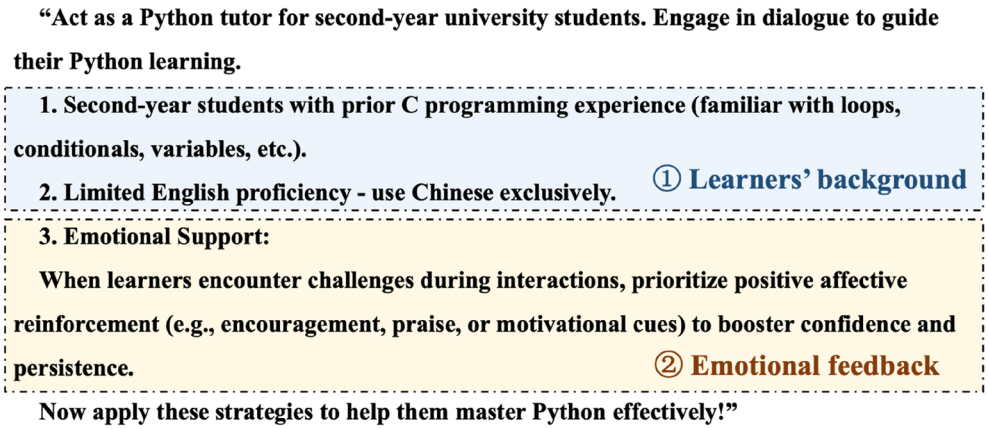


FIGURE 3 | Prompt design illustrating the core components of the Socratic-directed strategy.

TABLE 2 | Analytical framework.

Perspective	Data type	Data source	Analytical method	Research question
Process-oriented	Process data	Screen recordings	Click stream analysis (CsA)/Lag-sequential analysis (LsA)	How did the impact of GSL on learners' programming behaviours contrast with the effect of GDL?
	Platform data	Platform log data	Epistemic network analysis (ENA)	How did the impact of GSL on learners' interaction qualities with GAI contrast with the effect of GDL?
Summative	Perception data	Survey	Analysis of Covariance (ANCOVA)	How did the impact of GSL on learners' perceptions towards GAI contrast with the effect of GDL?

resulted in a total of 3480 min of recorded video data from our 80 participants, reflecting an actual average session length of 43.5 min per participant. This decision was driven by two factors: (1) the session centred on a comprehensive programming task, enabling observation of students' full problem-solving process with GPT integration and (2) it provided clearer insights into their programming proficiency. For analysis, click stream analysis techniques (Filvà et al. 2019) were applied to identify behavioural patterns.

The video data were analysed through an iterative coding process to identify student programming behaviours. For this purpose, we adopted the coding framework detailed in Table 3, which was adapted from a prior, validated study by Sun, Boudouaia, Zhu, et al. (2024). This framework's established validity for analysing student-AI interaction in programming contexts ensured its relevance to our study. To guarantee the reliability of the coding, a rigorous inter-rater reliability (IRR) procedure was conducted. Two trained coders, Ph.D. candidates

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TABLE 3 | Coding framework for programming behaviour.

Category	Code	Description
Python-related behaviours	Understanding Python codes (UPC)	A student attempted to understand the code with the mouse moving back and forth on the code.
	Coding in Python (CP)	A student wrote codes with the Python language in the system.
	Debugging in Python (DP)	A student debugged in PyCharm.
	Reading console message (RCM)	A student read error messages in output console in PyCharm.
GPT-related behaviours	Pasting website' codes (PWC)	A student pasted website' codes to GPT.
	Asking new questions (ANQ)	A student asked new questions in GPT.
	Asking further questions (AFQ)	A student asked questions to the feedback of GPT.
	Answering GPT's questions (AGQ)	A student answered questions from GPT.
	Pasting console message (PCM)	A student pasted error messages from output console window in GPT.
	Pasting Python codes (PPC)	A student pasted Python codes to GPT.
	Failure in GPT (FC)	GPT failed to give feedback due to technical problems (e.g., cannot connect to the server, get stuck).
	Reading feedback (RF)	A student read feedback in GPT.
	Copy and paste codes (CPC)	A student copy and paste codes from GPT.
Other behaviours	Referring to additional materials (RAM)	A student referred to additional materials from instructors.
	Other operation (OO)	A student opened VScode, installed third-party library, create new python file, checking website codes etc.
	Idle operation (IO)	A student had no operation.

in Educational Technology with prior experience in qualitative analysis of learning processes, independently analysed a randomly selected subset of 20% of the videos. The agreement between them was measured using Cohen's κ, yielding a coefficient of 0.89, indicating a strong level of agreement. Any discrepancies were resolved through discussion between the coders. After establishing this high reliability, one of the coders completed the analysis for the remaining video data. Independently of coding outcomes, lag-sequential analysis (Faraone and Dorfman 1987) was employed to examine behavioural transitions, including frequency patterns and visualised networks across teaching modes. Transitional association strength was quantified using Yule's Q, chosen for its capacity to account for baseline rates and descriptive clarity.

The second approach focused on analysing learners' interaction patterns using log data from the GPT Next platform, which tracked question submissions and feedback exchanges. Guided by the knowledge inquiry framework of Sun, Boudouaia, Zhu, et al. (2024), a model evaluating the complexity of student questions and their iterative engagement with AI feedback, raw log data were first organised in Excel, sorted by learner ID, performance level and query content. After systematic anonymization, interactions were divided into discrete coding units (461 for GSL; 369 for GDL) based on syntactic and semantic boundaries. Two trained researchers independently applied a predefined coding scheme (Table 4), achieving strong inter-rater reliability (0.83), which bolstered methodological rigour. Finally, coded data were converted into adjacency matrices for epistemic network analysis (ENA), a technique mapping relationship in knowledge-building behaviours. This enabled the use of webENA, an open-source tool, to visualise and compare shifts in inquiry strategies across performance groups.

The third component of the study assessed learners' perceptions of GPT-4 in both instructional groups through a post-course survey adapted from the Technology Acceptance Model (TAM) framework, which is used in previous studies (Sun, Boudouaia, Yang, et al. 2024; Sun, Boudouaia, Zhu, et al. 2024). The survey was divided into two parts: the first part gathered demographic information, prior technology exposure and learner expectations via open-ended and multiple-choice questions. The second part utilised a 5-point Likert scale (1 = *strongly disagree* to 5 = *strongly agree*) to assess four core TAM dimensions: perceived usefulness, ease of use, behavioural intention to adopt GPT-4 and attitudinal acceptance. The acceptance construct was specifically adapted to incorporate ethical concerns associated with GAI. Quantitative analysis employed descriptive statistics to identify trends in adoption readiness and ANCOVA to compare group differences in TAM scores, adhering to psychometric validation protocols.

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4 | Results

4.1 | College Students' Programming Behaviours in GSL and GDL Class

In addressing RQ1, we collected and observed the student activity during the programming process. To identify any potential divergences in behavioural patterns between the two instructional paradigms, we implemented a lag-sequential analysis. Notably, the behavioural patterns of learners from both groups illuminated a blend of similarities and distinct differences.

From a frequency perspective, if we set aside irrelevant behaviours, system operation in GAI, the GSL and GDL group both applied reading feedback (RF: GSL = 1643; GDL = 2180) frequently and understanding Python code (UPC: GSL = 610; GDL = 997). In contrast, GSL had much more behaviours in

TABLE 4 | Encoding framework for interaction with GPT-4.

Category	Code	Description	Sample
Knowledge inquiry (KI)	Irreverent knowledge inquiry (IKI)	Students raise questions unrelated to the topic.	What do you think could be improved about the current learning model in middle school?
	Superficial-level knowledge inquiry (SKI)	Students raise questions related to the topic, but without personal processing or perspectives	How to get the content on the web page?
	Medium-level knowledge inquiry (MKI)	Students raise questions related to the topic, which have been processed by themselves	I need a Python web crawler code that can scrape teacher information from the college website.
	Deep-level knowledge inquiry (DKI)	Students raise questions related to the topic, clearly articulate their viewpoints, arguments or opinions and provide detailed problem-solving steps	Please write a Python script using the requests, re and os libraries to perform web scraping on the webpage 'https://***'. The script should analyse the webpage structure, parse its content, extract information about Educational Technology faculty members and save the data
Feedback inquiry (FI)	Superficial-level feedback inquiry (SFI)	Student simply (dis)agreement or simple evaluation for the feedback from GPT-4.	I don't know.
	Medium-level feedback inquiry (MFI)	Students evaluate GPT-4's feedback and ask for more simple questions.	The page at https://*** could not be found. Could you help locate it or suggest alternative resources?
	Deep-level feedback inquiry (DFI)	Students evaluate the feedback on GPT-4 and provide specific examples for further information.	I have confirmed that both libraries are installed and VS Code has been restarted. However, the following error occurred: ModuleNotFoundError: No module named 'requests'

(Continues)

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TABLE 4 | (Continued)

Category	Code	Description	Sample
Feedback from GPT-4 (FC)	Task-based feedback (TbF)	GPT-4 summarises or outlines the student's question or points out issues within it.	This error message indicates that the faculty_list variable is None before attempting to use the .find_all ('a') method.
	Process-oriented feedback (PoF)	GPT-4 analyses the student's question step-by-step and directly provides specific code or solutions.	To provide the HTML structure of a webpage, follow these steps: (1) Open the webpage you want to view. (2) Right-click on the webpage and select 'View Page Source' (or find the corresponding option in the browser menu, usually in 'Developer Tools' or 'Tools').
	Heuristic-based feedback (HbF)	GPT-4 does not directly provide the answer but guides the student to think step-by-step and asks questions about their knowledge or skills.	But before that, I would like to ask you a few questions to help me better understand the problem: Have you confirmed that the faculty_list variable contains all the teacher information? Have you tried printing the faculty_links list to see how many elements it contains? We can start exploring from these questions and then gradually solve the problem.
	Affective feedback (AF)	GPT-4 provides feedback on the student's question that is complimentary, critical, supportive or empathetic.	Great job, you've started writing the code. If you run into any trouble or have any questions, let me know and I'll help you resolve them.
	Ethical feedback (EF)	GPT-4 remind students to pay attention to ethical issues	Warning: Scraping website data without authorization may violate the Terms of Service or laws and regulations. Always ensure that your crawlers comply with laws and regulations and the terms of use of the site.
	Incorrect feedback from GPT-4 (IFC)	GPT-4 provides incorrect feedback or is unaware of how to answer the student's question.	Sorry, due to copyright restrictions, I am unable to provide the corresponding answer.

answering questions from GPT-4 (AGQ: GSL = 360; GDL = 3), while the GDL group had dominant behaviours in asking new questions (ANQ; frequency = 967) (see Figures 4 and 5).

Meanwhile, regarding sequence analyses, both the GSL and GDL classes featured 54 and 48 significant programming learning sequences respectively, boasting ample interconnections across distinct codes (see Table 5). GSL students predominantly: (1) debugged in Python and read console messages to check the output (DP → RCM, Yule's $Q = 0.99$); (2) copied and pasted codes from GPT-4 to VScode and asked further questions from GPT-4 (PPC → AFQ, Yule's $Q = 0.96$); and read console messages and pasted these messages to GPT-4 to look for solutions (RCM → PCM, Yule's $Q = 0.90$). In contrast, GDL students mainly: (1) referred to additional materials to understand the Python codes (RAM → UPC, Yule's $Q = 0.99$); (2) had irrelevant behaviours to understand the Python codes (IO → UPC, Yule's $Q = 0.98$) and pasted Python codes into GPT-4 to ask further questions to the feedback from GPT-4 (PPC → AFQ, Yule's $Q = 0.96$). These variations highlight how distinct student cohorts adopt learning approaches tailored to their environmental conditions and resource availability.

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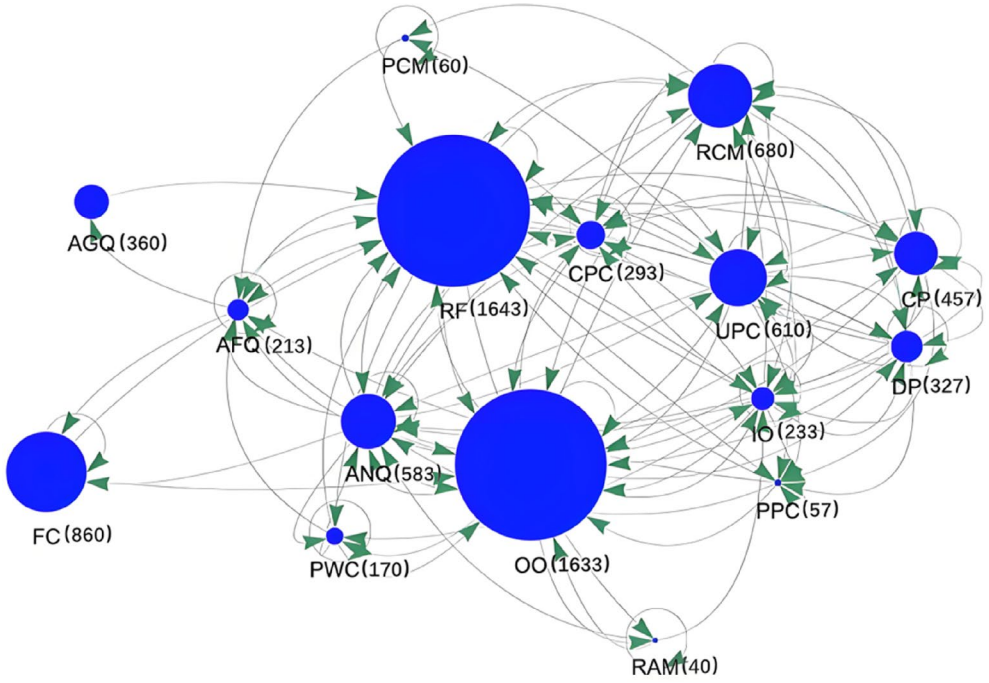


FIGURE 4 | Student's behavioural sequence diagram in GSL. The nodes represent behaviour codes, the number near each node is its frequency and arrows show the directional flow of the interaction, representing a significant transition from one behaviour to the next.

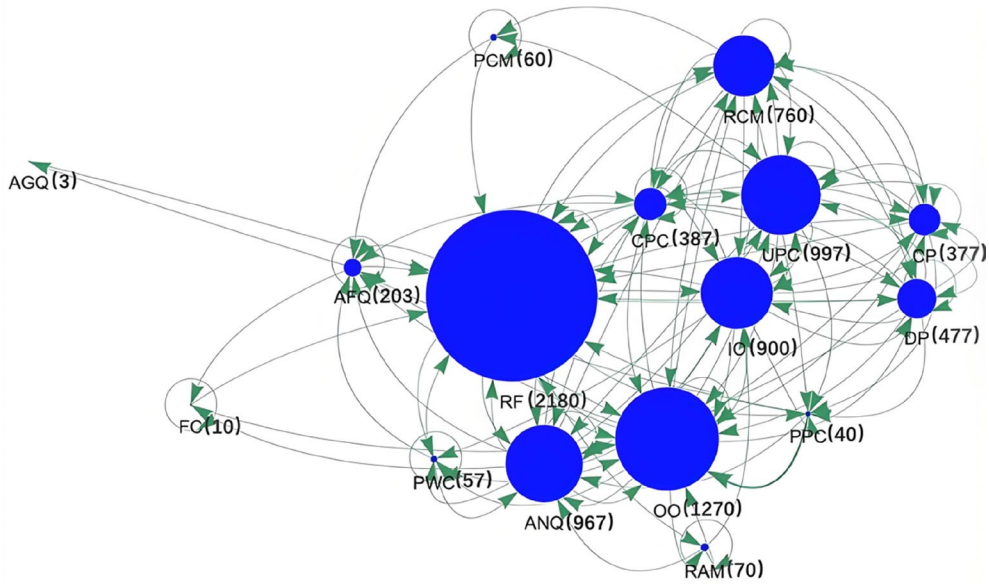


FIGURE 5 | Student's behavioural sequence diagram in GDL. The nodes represent behaviour codes, the number near each node is its frequency, arrows show the directional flow of the interaction, representing a significant transition from one behaviour to the next.

4.2 | College Students' Interaction Qualities With GPT-4 in GSL and GDL Class

Regarding RQ2, we collected data on students' interaction with GPT-4 in two groups and ENA was applied to discover differences in the cognitive level between the two groups. Figure 6a,b exhibit the epistemics networks derived from the discourse of the GSL and GDL groups. In these representations, nodes embody the codes, the lines interconnecting the nodes symbolise the corresponding connections and line thickness is indicative of the relative co-occurrence frequency (Shaffer et al. 2016). Based on the interaction pattern analysis between knowledge inquiry levels and GPT-4 feedback types, the study reveals distinct learning pathways: Students under the GSL methodology depicted strong connections in AF-PoF (0.36), PoF-HbF (0.26), AF-HbF (0.23), PoF-SFI (0.19) and MFI-HbF (0.17). On the other hand, GDL students represented potent connections in SKI-PoF (0.27), SKI-MKI (0.22), MKI-PoF (0.21), PoF-EF (0.19), SKI-EF (0.19), SKI-SFI (0.19) and SKI-AF (0.19). GSL methodology fostered an affectively supported learning cycle, evidenced by strong AF-PoF linkages where affective feedback amplified engagement with

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process-oriented guidance and synergistic PoF-HbF connections and AF-HbF correlation indicating heuristic prompts effectively promoted step-by-step instructions and emotional reinforcement. Similarly, GDL methodology exhibited dominant process-oriented feedback from GPT-4 (SKI-PoF), reflecting GPT-4 could offer procedural solutions for surface-level questions in general, while weaker SKI-EF links imply ethical awareness rarely emerged without affective scaffolding. GSL's seem to have advantage in cultivating ethical-technical integration through strong connections in AF-PoF.

The subtracted network model effectively captures distinct categorical association patterns between the two learner clusters (Sun et al. 2022). Figure 7 visualises the differential interaction patterns with GPT-4 through dimensional reduction analysis, where the first principal component (*X*-axis) explains 29.2% of variance and the second component (*Y*-axis) accounts for an additional 13.8%. Independent samples *t*-tests revealed significant between-group divergence along the primary dimension (MR1), with GSL students showing substantially higher coordinate values ($M = 0.28$, SD = 0.18) compared to GDL peers ($M = -0.40$, SD = 0.13). This statistically robust differentiation ($t(31.98) = -12.58$, $p = 0.0000$, Cohen's $d = 4.14$) along the primary axis underscores fundamentally distinct human-AI interaction modalities. For context, while a Cohen's *d* of 0.8 is typically considered a large effect, the value of 4.14 observed here is exceptionally large. It indicates that the distributions of communicative behaviours for the GSL and GDL groups are so distinct as to be practically non-overlapping, reinforcing the profound impact of the pedagogical intervention. Compared to students learning in the GDL group, those in the GSL group demonstrated enhanced interconnections between PoF-HbF (0.24), MFI-PoF (0.21), MFI-AF (0.20) and PoF-AF (0.19) links than their GDL counterparts. Conversely, the GDL students exhibited stronger SKI-MKI (−0.18), SKI-EF (−0.16) and SKI-SFI (−0.14) links than their GSL counterparts.

TABLE 5 | LsA transition frequency of programming behaviours of learners in two groups.

GSL		GDL	
Transition	**Yule's *Q***	**Transition**	**Yule's *Q***
DP-RCM	0.99	RAM-UPC	0.99
PPC-AFQ	0.96	IO-UPC	0.98
RCM-PCM	0.90	PPC-AFQ	0.96
CP-DP	0.87	RCM-PCM	0.95
PCM-AFQ	0.84	DP-RCM	0.95

4.3 | College Students' Perceptions Towards GPT-4 in GSL and GDL Class

Concerning RQ3, we collected student feedback through surveys and employed a paired *t*-test to examine perceptions towards GPT-4. In regard to pre- and post-test changes of perceptions towards GPT-4 between two groups, the results revealed (see Table 6) that students in both groups experienced a significant improvement in the aspect of perceived ease of use (GSL: $t = -2.76$, $p = 0.007$; GDL: $t = -2.90$, $p = 0.006$) and intention to use (GSL: $t = -2.02$, $p = 0.047$; GDL: $t = -2.01$, $p = 0.047$) of GPT-4, while students in the GDL group demonstrated a declined trend in the attitude of using GPT-4 ($t = 1.12$, $p = 0.269$).

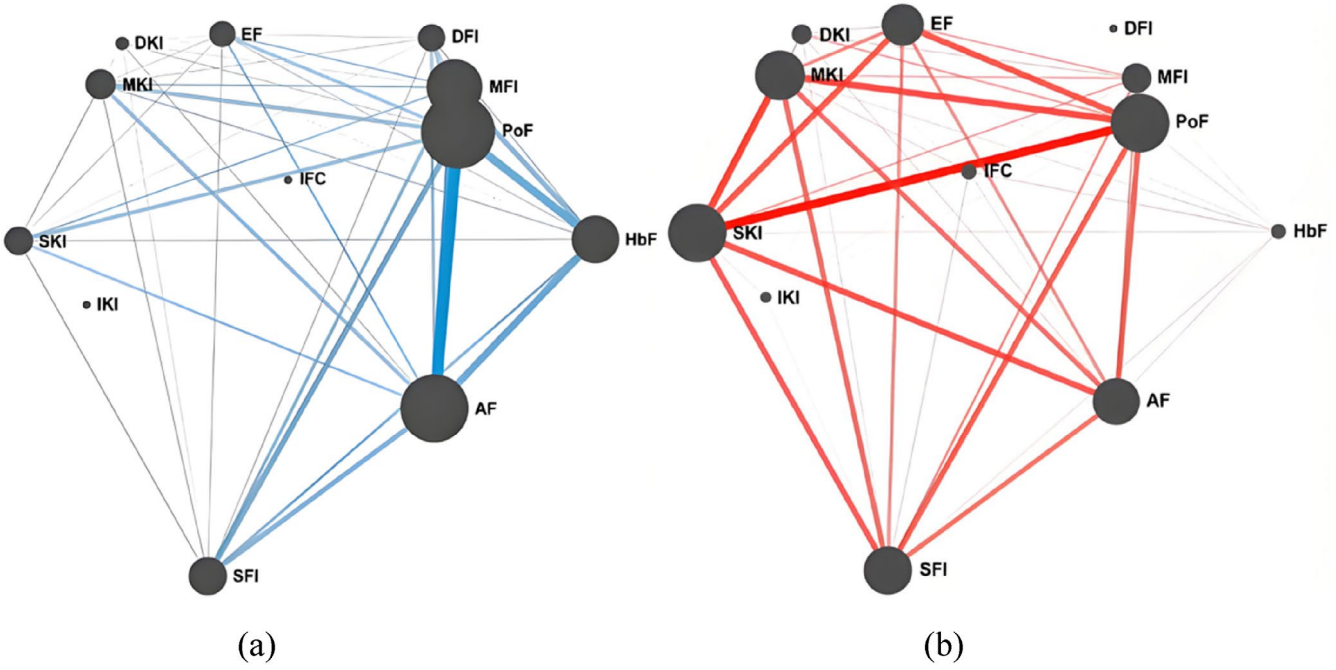


FIGURE 6 | Students' cognitive network of (a) GSL (in blue) and (b) GDL (in red). The nodes represent the key cognitive codes identified during analysis and the size of the nodes represent the frequency of the codes. The lines indicate the co-occurrence of two codes, with the line thickness being proportional to the strength of the connection.

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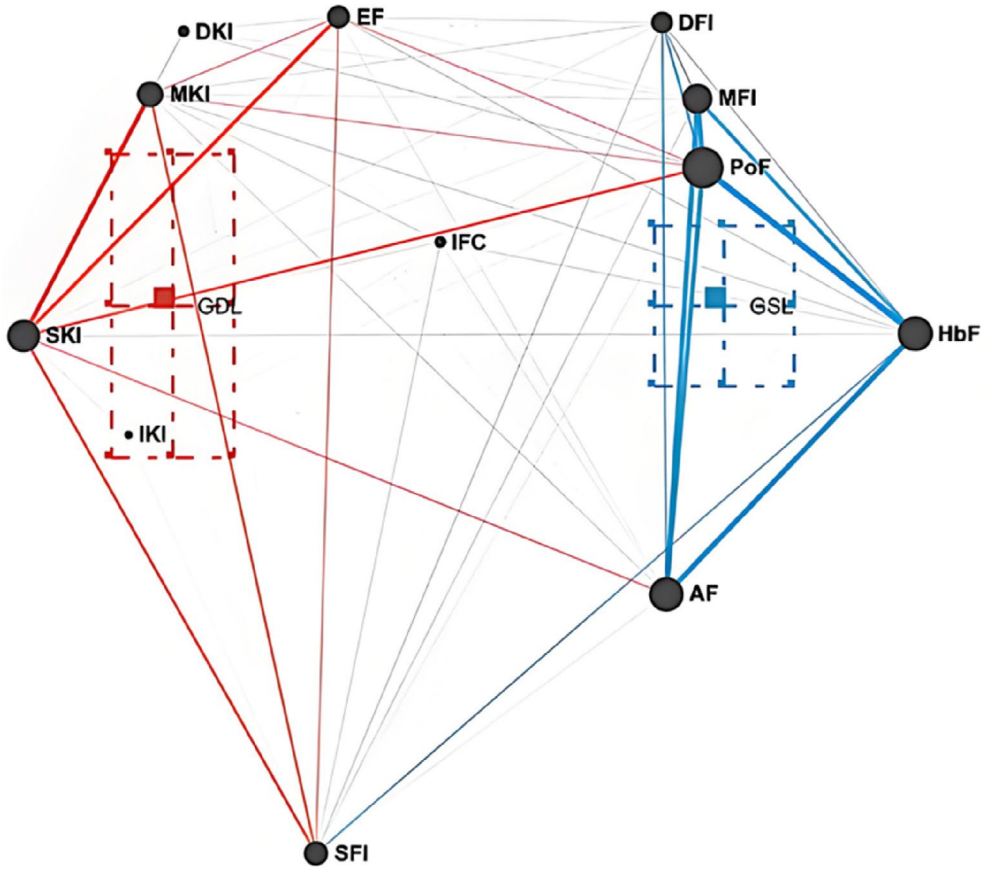


FIGURE 7 | Subtracted network of interaction level of GSL (in blue) and GDL (in red). The nodes represent the key cognitive codes identified during analysis and the size of the nodes represent the frequency of the codes. The lines indicate the co-occurrence of two codes, with the line thickness being proportional to the strength of the connection. This subtraction plot highlights the differences between the groups: blue edges represent connections that were significantly stronger in the GSL group, while red edges represent connections that were significantly stronger in the GDL group.

5 | Discussion

This study investigated how the Socratic questioning method integrated in GAI influences college students' programming behaviours, interaction qualities and perceptions by systematically comparing GSL and GDL. The results revealed both overlapping and distinct behavioural patterns between the groups, with key differences largely driven by the different questioning method.

Firstly, the deployment of lag-sequential analysis allowed for an examination of the students' activities during the programming process, revealing distinct patterns of behaviour reflective of the two learning methods. In terms of frequency, both GSL and GDL students demonstrated frequent engagement with reading GAI-generated feedback and exhibited active efforts to understand Python code. However, distinct behavioural patterns emerged. GSL students prioritised reading console messages and responding to GAI's scaffolded questions, reflecting the efficacy of Socratic questioning in guiding learners through iterative, step-by-step problem-solving (Chang 2024). In contrast, GDL students more frequently posed new questions to GAI, aligning with its inherent capacity for open-ended dialogue (Gayed et al. 2022). Notably, GDL learners also engaged in higher rates of Python code adaptation, suggesting a trial-and-error approach. However, their reliance on copying and modifying GAI-generated code raises concerns about over-dependence on external solutions, which may hinder the development of independent problem-solving skills (Krupp et al. 2024). The sequence analysis further differentiates the learning behaviours between GSL and GDL. GSL students, guided by Socratic questioning method, exhibited a 'cyclical, reflective approach' to programming. Their dominant behavioural sequences, such as debugging Python code, pasting code to VScode followed by iterative questioning and leveraging console messages to troubleshoot errors, reflect a structured, process-oriented engagement, suggesting that scaffolded prompts fostered a methodical workflow akin to expert problem-solving (Sun, Boudouaia, Yang, et al. 2024; Sun, Boudouaia, Zhu, et al. 2024). Conversely, GDL students prioritised 'expedient solutions' over systematic inquiry. Their reliance on external materials and frequent code pasting to solicit direct answers highlights a dependency on external resources rather than internalised reasoning. GDL students demonstrated adaptability in navigating GAI's open-ended interface, their tendency to bypass diagnostic steps (e.g., debugging) in favour of immediate code adjustments risks fostering 'procedural mimicry' over genuine computational thinking. This aligns with prior critiques of AI tools in education, where overreliance on automated solutions can undermine skill transfer and independent problem-solving (Amos et al. 2025; Yilmaz and Karaoglan Yilmaz 2023).

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TABLE 6 | Paired samples *t*-test results of pre-test and post-test of GSL and GDL.

Dimension	Group	Test	*M*	SD	df	*t*	*p*
Perceived usefulness	GSL	Pre-test	3.87	0.54	70	−1.63	0.107
		Post-test	4.12	0.76			
	GDL	Pre-test	4.18	0.58	86	−0.90	0.369
		Post-test	4.29	0.56			
Perceived ease of use	GSL	Pre-test	3.73	0.61	70	−2.76**	0.007
		Post-test	4.17	0.73			
	GDL	Pre-test	3.93	0.68	86	−2.90**	0.006
		Post-test	4.19	0.73			
Intention to use	GSL	Pre-test	3.85	0.65	70	−2.02*	0.047
		Post-test	4.19	0.73			
	GDL	Pre-test	4.30	0.67	86	−2.01*	0.047
		Post-test	4.55	0.49			
Attitude	GSL	Pre-test	3.75	0.65	70	−0.38	0.701
		Post-test	3.82	0.80			
	GDL	Pre-test	4.22	0.68	86	1.12	0.269
		Post-test	4.06	0.60			

*$p < 0.05$.
**$p < 0.01$.

Second, the ENA findings suggest that the GSL approach fosters deeper cognitive engagement and emotional support, as evidenced by the stronger connections between heuristic-based feedback and different level of inquiries and stronger connections between affective feedback and process-oriented feedback. These linkages suggest that Socratic questioning strategies fostered an iterative cycle where emotional reinforcement (e.g., encouragement, motivation) amplified learners' engagement with step-by-step problem-solving. For instance, when students encountered errors, affective feedback from GPT likely reduced frustration, enabling them to persist with debugging and refine their code using heuristic guidance. This aligns with constructivist principles, where scaffolding that integrates metacognitive and emotional support promotes deeper conceptual understanding through active knowledge application (Held et al. 2024). The significant AF-PoF-HbF triad in GSL reflects a learning environment where learners dynamically connected technical problem-solving with self-regulated reflection, mirroring expert-like cognitive practices (Sun et al. 2022). In contrast, GDL students demonstrated dominant surface knowledge inquiry (SKI) interactions, such as SKI-PoF and SKI-MKI, which reflect a tendency to seek immediate solutions rather than engage in diagnostic reasoning. The prevalence of SKI-EF ethical feedback and SKI-SFI links further underscores the absence of affective or heuristic scaffolding in GDL. Without guided inquiry, learners defaulted to copying code and soliciting direct answers, resulting in fragmented knowledge networks. Our study demonstrates that GSL environments promote an interaction dynamic that encourages learners to engage in more meaningful and complex discourse with GPT, thereby potentially enhancing their understanding and retention of the material.

The subtracted network and the significant differences found in the comparative analysis show that the GSL and GDL groups communicate in very different ways. This finding is consistent with the work of Sun et al. (2022), who found that the structure of learning activities significantly influences the quality of student interactions with learning technologies. Moreover, the enhanced interconnections between mid-level knowledge indicators in the GSL group suggest that this approach facilitates a more integrated understanding of concepts, as learners are more frequently connecting different levels of knowledge. This observation supports the assertion by Shaffer et al. (2016) that the structure of knowledge connections within a learning environment can impact the depth of student understanding and learning outcomes.

Third, the analysis of learners' perception revealed a notable contrast between GSL and GDL learners' perceptions of GAI. While both groups reported significant improvements in 'perceived ease of use' and 'intention to use', students in the GDL group exhibited a decline in their 'attitude towards using GAI' post-intervention. This divergence can be interpreted through the lens of the TAM, which posits that attitude is a critical mediator between perceived ease of use and the ultimate intention to use a technology (Davis 1989). Our findings suggest that instructional design is a key external variable influencing this entire process. And this finding also resonates with the work of Tu and Hwang (2023), who found that university students' conceptions of GAI differ based on their achievement levels; high-achieving learners tend to perceive ChatGPT as a 'tutor' for higher-order cognitive tasks like creating, whereas low-achieving learners view it as a 'tool' for lower-order tasks such as

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remembering and understanding. In our study, for GSL learners, the GSL framework's structured prompts likely aligned learners' expectations with GPT's capabilities, fostering satisfaction and sustained positive attitudes. The scaffolding provided by GSL may have reduced extraneous cognitive load, enabling learners to focus their mental resources on understanding the programming concepts rather than on how to formulate effective queries (Sweller 1994). By contrast, the GDL group's decline in attitude suggests that unstructured or self-directed interactions with GAI, lacking explicit guidance, may have induced a high cognitive load, leading to frustration and mismatched expectations (Kirschner et al. 2006) and lead learners to default to a 'tool' orientation. This limited perception, when misaligned with complex learning expectations, can result in frustration or mismatched expectations, hence the negative shift in attitude. This supports McTear et al.'s (2023) assertion that learners benefit most when supported by pedagogical frameworks that scaffold understanding. The GSL group's preference for structured, reflective prompts highlights the importance of clarity and active knowledge construction in GAI-mediated learning. Learners sought not just answers but a coherent, logical process, reinforcing the idea that effective GAI integration requires deliberate instructional strategies to balance autonomy with guidance.

6 | Implications

Building on these findings, this study proposes pedagogical frameworks emphasising guided scaffolding and cognitive partnership with GAI, alongside developmental strategies to equip educators with structured prompt design, Socratic questioning and iterative feedback techniques. Pedagogically, integrating Socratic questioning into GAI fosters structured, reflective engagement, aligning with constructivist principles. First, the findings synthesise into a clear imperative for structured pedagogical integration. The observed decline in attitudes among GDL learners is not merely a finding to be reported; it is a direct signal of the pitfalls of unguided GAI use, where a mismatch between expectations and capabilities can lead to disillusionment (Liu et al. 2023). Therefore, the primary implication for instructional design is the need for a systematic onboarding process. Educators must proactively train learners on how to engage with GAI as a thinking partner. This involves teaching structured prompt design, iterative feedback techniques and metacognitive skills like evaluating the AI's responses, thereby framing the tool's limitations as pedagogical opportunities for critical thinking. Second, effective pedagogy requires that Socratic questioning be adapted to the specific programming context. A one-size-fits-all prompting strategy is suboptimal. For instance, when a learner is struggling with a syntax or runtime error, the Socratic prompts should be highly targeted to guide their attention and diagnostic reasoning (e.g., 'The error message mentions a "TypeError" on line 10. Can you explain what data types you are trying to combine in that line?'). In contrast, when a learner faces a deeper logical or conceptual challenge, the prompts should foster broader metacognitive reflection. In this context, questions like, 'Could you explain the reasoning of your algorithm step-by-step?' or 'What are the potential gaps or edge cases in the AI's suggested solution?' are more effective. This context-aware adaptation emulates expert-like problem-solving, manages cognitive load and shifts the learner's focus from immediate answer retrieval to a more profound cognitive partnership.

From a developmental perspective, our findings underscore the risk of GAI tools stifling independent thinking if they primarily offer direct solutions. To cultivate transferable skills, future educational technologies should be designed with the following in mind. First, GAI systems should be architected to provide adaptive scaffolding that evolves with a learner's proficiency. Instead of static responses, the AI could gradually decrease the specificity of its hints as a learner demonstrates competence, thus encouraging autonomy. This aligns with developmental goals of fostering resilience and self-efficacy. Second, drawing from recent work (Chang 2024; Etkin et al. 2025), GAI tools should be enhanced to offer not just cognitive but also heuristic and affective support. The system could be designed to prioritise guiding questions over direct code generation and, potentially, to recognise signs of learner frustration and respond with encouragement or a shift in strategy. This would help bridge the gap between technical execution and conceptual understanding. By aligning GAI capabilities with these pedagogical and developmental goals, we can create learning environments that foster holistic skills. Future research should continue to explore hybrid models that merge structured Socratic guidance with opportunities for open-ended exploration, thereby optimising GAI-augmented learning for both structured skill acquisition and creative problem-solving.

7 | Conclusion, Limitations and Future Directions

GPT-4, an advanced AI language model, demonstrates significant potential to enhance computer programming education by offering guidance, answering questions and generating code snippets. This quasi-experimental study compared two pedagogical approaches, namely GSL and GDL, to analyse their effects on college students' programming behaviours, interaction qualities and perceptions of GAI. The findings indicate that integrating GAI tools like GPT into programming curricula is feasible, particularly when structured strategies, such as the Socratic questioning method, are incorporated to optimise interactions with GAI.

While this study provides valuable initial insights into GAI-supported programming education, its findings should be interpreted considering several limitations. These limitations also highlight important avenues for future research. A primary concern is the potential for instructor bias, as the first author's dual role as researcher and instructor for both groups may have unintentionally influenced student engagement. The study's external validity is also constrained by its context, a single university and a single programming language (Python), meaning the generalizability of our GAI-scaffolding model requires replication across diverse settings and languages. Furthermore, we did not formally control for students' prior programming experience, which likely moderates how learners with different skill levels benefit from GAI interaction. Finally, the study is limited by its technological design, using a static version of GPT-4 and a foundational Socratic questioning method. Given the rapid pace of AI development, future work must evaluate newer models and,

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more importantly, develop sophisticated, adaptive systems that can dynamically tailor scaffolding to individual learner trajectories, moving towards a true cognitive partnership. In particular, integrating a knowledge graph, as effectively demonstrated by the IntelliChain model (Qi et al. 2024) in mathematics, represents a critical path to achieve this within programming education. Despite these constraints, the results suggest that strategically designed Socratic questioning method can significantly improve the educational utility of GAI. Further academic inquiry is critical to expand on these findings and comprehensively evaluate how AI language models can optimise programming instruction. Ultimately, tailored GAI-based strategies may serve as powerful pedagogical tools, enhancing learning experiences for educators and students alike in computer programming education.

Author Contributions

Dan Sun: conceptualization, investigation, methodology, formal analysis, writing – original draft, writing – review and editing. **Yi Zheng:** writing – review and editing. **Jie Xu:** validation, writing – review and editing. **Zhanshan Yang:** writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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